

# JONATHAN ARMSTRONG

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## SUMMARY

ML and AI engineer who builds and ships end-to-end. Self-taught alongside a full-time operations role, I have shipped a stack of Model Context Protocol (MCP) servers now live on the official MCP registry and PyPI, a foundation reinforcement-learning model for workforce scheduling, and internal automation I run day to day. Strong with PyTorch, PPO, transformer architectures, and agentic LLM workflows. Looking for a role in applied AI, forward-deployed engineering, or ML engineering where the job is to build things that ship.

## CORE SKILLS

**AI/ML:** PyTorch, PPO, LSTM, custom transformer (multi-head self/cross-attention, actor-critic), QLoRA/LoRA fine-tuning, curriculum learning, batched RL training, CUDA

**LLM/Agentic:** MCP protocol (published to the official registry), A2A protocol, agentic workflows, Claude API, JSON-RPC stdio servers, local LLMs (llama.cpp, Ollama)

**Backend/Infra:** FastAPI, Flask, Pydantic, REST APIs, SQLite, Redis, APScheduler, Docker, Cloudflare Workers and D1, GitHub Actions

**Frontend/Data:** React 19, Supabase, Vite, Tailwind, pytest, pandas, plotly

**Automation:** Python (Selenium, openpyxl), NetSuite, ShipExec, Amazon Seller Central, CommerceHub, Magento

## PORTFOLIO

### Xylem

AI agent infrastructure, live on the official MCP registry and PyPI. A local-first stack of MCP servers for multi-agent AI development, built on the insight that git push is a compare-and-swap. context-keeper (rationale-first memory, ~92% context reduction with calibrated abstention), agentsync (server-less coordination, 1,000-race concurrency test with zero lost or double-granted claims), and cambium (knowledge lifecycle: distill, recall, promote). Cloudflare Worker remotes make a phone a full peer.

### Clark

Foundation RL model for warehouse workforce scheduling and the capstone of a self-built progression (Dolly to Jack to Clark). Transformer plus LSTM, 18M params, IPPO per-worker ratio clipping, symlog value targets, curriculum across 5 to 50 worker facilities. Holds ~93% order completion uniformly across facility sizes. FastAPI multi-facility API with auth, Docker, and a pre-train / fine-tune pipeline.

### Waveform-MCP

The only MCP server for Tracktion Waveform. ~150 tools, 8 generative composers, audio analysis (LUFS, spectrum, tempo, key), BasicPitch transcription, plugin-aware fader calibration, and a reverse-engineered .tracktionedit format. 250-plus tests.

### Balatron

PPO agent that plays Balatro, an active project still training today. 833-dimensional state vector encoding joker fingerprints, economy state, hand evaluation, and boss-blind strategy. 17-plus confirmed white-stake wins. PyTorch, CUDA.

### Skein

Local-first observability for the A2A protocol. Six-tool MCP server, eight-page dashboard, 71 tests, CI across three Python versions. Empty repo to shipped tool in under two hours.

## EXPERIENCE

### Operations Manager and Internal Automation Engineer | VOLT / LLW Enterprise

March 2020 to Present

- Run a 6-person warehouse team in a 200-plus order per day environment; track KPIs across shipping, inventory, and labor; lead weekly 1:1s and bi-weekly team check-ins.
- Built MasterMetrics (Python, Selenium) as an internal proof of concept. Measured time savings project to roughly \$17K per year per location in reporting labor. I run it day to day for my own team.
- Built a bin-label generator (Python, openpyxl, reportlab) that turns an Excel pick-list into print-ready PDF labels with scannable Code 39 barcodes. Open-sourced as barcode-label-maker.
- Serve as the informal IT and automation resource for leadership across NetSuite, ShipExec, Amazon Seller Central, CommerceHub, and Magento.
- Self-taught the entire AI/ML portfolio above outside of work, on nights and weekends.

## EDUCATION

### Associate Degree, Computer Networking and Internet Technologies | Keystone Technical Institute, Harrisburg PA

January 2014

Magna Cum Laude | GPA 3.8